Sushruta Behera

Master of Technology
in Computer Science and Engineering
Indian Institute Of Technology, Ropar

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GitHub
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Education

Degree	Institute/Board	CGPA/Percentage	Year
Master of Technology	Indian Institute of Technology, Ropar	8.71 (Till 3rd Sem)	2024-2026
Bachelor of Technology	Veer Surendra Sai University of Technology	8.77	2019-2023
Senior Secondary	Council of Higher Secondary Education, Odisha	72.5%	2018
Secondary	Board of Secondary Education, Odisha	92.83%	2016

Projects

- **VARTA – Real-Time Chat App** *April 2025 - June 2025*
  Real-Time Chat Application **Github**
  - Developed a real-time chat app using MERN stack and Socket.io for instant messaging
  - Implemented JWT-based authentication and protected user sessions
  - Integrated Cloudinary for media uploads and managed responsive UI with Tailwind CSS and Zustand
  - Designed scalable project structure with separate frontend and backend deployments and environment setup
- **IOT-RADIO-SECURITY-SDR** *Mar 2025 - Apr 2025*
  Security Research for 433 MHz IoT Networks **Github**
  - Built a testbed using ESP32 microcontrollers and RF modules to demonstrate vulnerabilities in IoT radio networks.
  - Performed replay attacks with ADALM-Pluto SDR and Universal Radio Hacker tools
  - Implemented sequence number-based mitigation achieving full prevention of replay attacks
  - Documented methodology and developed Arduino code with secure communication using RadioHead library
- **VIRTUAL-MEMORY-SIMULATOR** *Sep 2024 - Oct 2024*
  Memory Management and Paging Simulator **Github**
  - Built a C++ based simulator for virtual memory management with support for Map, Single-Level and Two-Level Page Tables
  - Implemented a multi-threaded trace generator using POSIX threads to create large memory access traces efficiently
  - Designed a configurable system via config.h, allowing flexible setup of memory layouts, page sizes and input trace file paths
  - Developed a performance analysis module to track hit/miss ratios, execution speed and memory allocation statistics
  - Automated build and execution with a single Makefile, supporting separate builds for simulator and trace generator
- **EasyMart - An E-commerce website** *2023*
  E-commerce Website **Github**
  - Built a feature-rich e-commerce site using Node.js and Express.js, handling all CRUD operations for products
  - Integrated Stripe for secure payment processing and implemented a robust cart management system
  - Managed user authentication, including signup with a welcome email via Nodemailer and password reset functionality
  - Employed MongoDB for data storage and EJS for a user-friendly frontend templating engine

Technical Skills

- **Programming Languages**: C, C++, Python, Javascript
- **Web Technologies & Frameworks**: Express, React.js, Node.js, RESTful APIs, WebSockets
- **Tools & Platforms**: Git, GitHub, VSCode, Docker, Linux, Shell Scripting
- **Databases**: MySQL, MongoDB

KEY COURSES TAKEN

- **CSE**: Operating System, Database Management System, Object Oriented Programming, Optimization for Machine Learning, Data Structures and Algorithm, Machine Learning, Artificial Intelligence
- **Others**: Computer System, Mathematical Foundation of Computer Science, Cybersecurity Essentials

POSITIONS OF RESPONSIBILITY

- **Teaching Assistant**: Summer Semester, IIT Ropar
- **Teaching Assistant, 2025**: First Semester, IIT Ropar

MISCELLANEOUS

- **Graduate Aptitude Test in Engineering 2024**: Computer Science and Information Technology(CS) Rank-1406